

DAX LAYER · VALIDATION · DISPLAY FOLDERS

Ride Booking Measure Layer

Forty-six measures across ten groups, built into the model and reconciled line by line against the raw 150,000-row file. Every KPI below has been executed, not just written.

Model NCR Ride Booking Executive Analytics Dashboard Home table Fact_Rides

Status created and verified

46
MEASURES

10
DISPLAY FOLDERS

11/11
KPIs RECONCILED

0
DOUBLE-COUNT RISKS

5
CORRECTIONS MADE

FOUNDATION

Three ways to count a booking — only one is right

Your brief asked this question explicitly, and on this dataset it is not academic. The three methods give three different answers, and the most commonly taught one is wrong here.

METHOD	DAX	RETURNS	ASSESSMENT
Count of rows	COUNTROWS (Fact_Rides)	150,000	CORRECT The grain is c per booking, guaranteed l unique Ride Key .Nothin be null, so nothing can be dropped.
Count of Booking	COUNT (Fact_Rides[Booking ID])	150,000	RIGHT ANSWER, WRONG It adrees today only beac

Booking IDs		it agrees today only because it is null. It counts <i>populated</i> not bookings — a single row a future refresh would undo the business with no error	
Distinct Booking IDs	DISTINCTCOUNT (Fact_Rides[Booking ID])	148,767	WRONG Loses 1,233 records. The audit proved 1,224 Booking IDs are reused across generations of different trips. This is the choice in most tutorials and would have quietly cost you a portion of every volume figure on the dashboard.

ADOPTED

Total Bookings = COUNTROWS (Fact_Rides). Every other volume measure in the model is a filtered call to this one, so the counting method is defined in exactly one place.

DISTINCTCOUNT is still the right tool for Customer ID — that column repeats *by design*, and distinctness is the question being asked. The rule is not “never use DISTINCTCOUNT”; it is “use it only where repetition is meaningful.”

FOUNDATION

Filter semantics: KEEPFILTERS, not plain CALCULATE

Every status-based measure — Completed Rides, Cancelled by Driver, and the rest — wraps its predicate in KEEPFILTERS . This is a deliberate choice with a visible consequence, so it is worth being explicit about.

Plain CALCULATE **REJECTED**

CALCULATE([Total Bookings], Dim_Booking_Status[Booking Status]="Completed") **replaces** any existing filter on that column. Click “Cancelled by Driver” on the status pie and the Completed

KEEPFILTERS **ADOPTED**

KEEPFILTERS(...) **intersects** with the incoming filter. Completed Rides is a *slice* of Total Bookings, and a slice must obey the same filters as the whole. Selecting a cancelled slice returns

Rides card still reads 93,000, ignoring the selection. The parts stop summing to the whole.

0 — honest, and arithmetically consistent.

Verified. Under a Vehicle Type = Auto filter, the five status measures sum to 37,419, which is exactly Total Bookings for that vehicle. Under no filter they sum to 150,000. The decomposition holds in every filter context tested.

CATALOGUE **The 46 measures**

Grouped exactly as the display folders are named, so what you read here is what you see in the Fields pane. All 46 live on Fact_Rides .

A Core KPI 5 MEASURES

The header row of the Overall Performance page. Everything else in the model is derived from these five.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS USED
Total Bookings	150,000	Every booking attempt, whatever the outcome. The demand signal, and the denominator for every rate in the model.	KPI card; Ride Volume Over Time; the base all volume visu
Completed Rides	93,000	Bookings that ended in a delivered trip. The fulfilment signal.	KPI card; Book Status Breakd
Completion Rate	62.0%	Share of all attempts that were fulfilled. The single number an executive checks first.	KPI card; trend by month; slice by vehicle
Total Booking Value	₹51,846,183	Gross billed value. Cancelled and unfulfilled bookings contribute nothing automatically — their value is null and SUM ignores nulls.	KPI card; Revenue by Payment Method; revenue trend
Total Ride Distance	2,512,975.2 km	Kilometres actually covered. The operational coverage signal.	KPI card; Top Vehicle Types Ride Distance

```
Total Bookings      = COUNTROWS ( Fact_Rides )

Completed Rides     = CALCULATE (
                      [Total Bookings],
                      KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "Completed" )
                    )

Completion Rate      = DIVIDE ( [Completed Rides], [Total Bookings] )

Total Booking Value  = SUM ( Fact_Rides[Booking Value (INR)] )

Total Ride Distance  = SUM ( Fact_Rides[Ride Distance (km)] )
```

B Booking

7 MEASURES

The status decomposition. The first five are mutually exclusive and exhaustive; the last two are business rollups that answer specific operational questions.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS USED
Cancelled by Customer	10,500	Rider-initiated aborts. Demand-side friction.	Status pie; Cancellation Intelligence page
Cancelled by Driver	27,000	Driver-initiated aborts. Supply-side friction, and by far the largest failure mode at 18% of all bookings.	Status pie; Cancellation Intelligence page
No Driver Found	10,500	Requests no driver ever accepted. Pure supply shortfall — nobody cancelled anything.	Status pie; supply KPI card
Incomplete Rides	9,000	Trips that started, were billed, and did not finish. Revenue-bearing failures.	Status pie; revenue reconciliation
Cancellation Count	37,500	Both cancellation types combined, via Status Group = "Cancelled" . Deliberately excludes No Driver Found.	Cancellation KPI card
Operational Rides	139,500	Bookings where a driver was actually allocated. The correct denominator for any	Driver Allocation Rate: operationa

		driver-performance metric.	denominators
Billed Rides	102,000	Bookings that generated an invoice (Completed + Incomplete). The correct denominator for every monetary and distance average.	Footnote on revenue cards; validation

The first four plus Completed Rides sum to exactly 150,000 — verified with and without slicers applied.



```
Cancelled by Customer = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "Cancelled by Customer" ) )

Cancelled by Driver    = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "Cancelled by Driver" ) )

No Driver Found       = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "No Driver Found" ) )

Incomplete Rides      = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "Incomplete" ) )

Cancellation Count    = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Status Group] = "Cancelled" ) )

Operational Rides     = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Booking Status] <> "No Driver Found" ) )

Billed Rides          = CALCULATE ( [Total Bookings],
    KEEPFILTERS ( Dim_Booking_Status[Is Revenue Generating] = TRUE ) )
```



Note how three of these read attributes off the dimension rather than re-listing status strings. Status Group and Is Revenue Generating were built in Step 03 precisely so this logic lives in one governed place. Change the classification of a status and every dependent measure follows.

c Revenue

5 MEASURES

Money. The one thing to get right here is the denominator on the averages.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WH
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Completed Ride Revenue	₹47,260,574	Realised revenue from delivered trips. Separating this from gross booking value quantifies exactly what incomplete trips are worth.	Revenue
Average Booking Value	₹508.30	Mean fare across the 102,000 billed rides. AVERAGE ignores blanks, so the denominator is correct without any filter.	KPI
Average Completed Ride Value	₹508.18	Mean fare of delivered trips only. Nearly identical to the above, which tells you incomplete trips are priced normally.	Revenue
Revenue per KM	₹20.63	Yield. The cleanest way to compare pricing efficiency across vehicle types, where raw revenue just tracks volume.	Revenue
Revenue Share %	contextual	Each row's share of the visible total, via ALLSELECTED . Respects slicers, so the shares always sum to 100% of what the user is looking at.	Revenue

```
Completed Ride Revenue      = CALCULATE ( [Total Booking Value],
      KEEPFILTERS ( Dim_Booking_Status[Booking Status] = "Completed" ) )

Average Booking Value      = AVERAGE ( Fact_Rides[Booking Value (INR)] )

Average Completed Ride Value = DIVIDE ( [Completed Ride Revenue], [Completed Rides]

Revenue per KM              = DIVIDE ( [Total Booking Value], [Total Ride Distance]

Revenue Share %             = DIVIDE (
      [Total Booking Value],
      CALCULATE ( [Total Booking Value], ALLSELECTED (
      )
      )
```

D Distance

2 MEASURES

Small group by design. Total Ride Distance already sits in Core KPI, and slicing it by Dim_Vehicle is what the Vehicle Type view needs — no extra measure required.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS
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Average Ride Distance	24.64 km	Mean trip length across billed rides. Establishes the typical journey and anchors the histogram's centre.	KPI card; Ride Distance Distribution; vehicle
Distance Share %	contextual	Each row's share of visible distance. Turns the Top 5 vehicle table from a raw ranking into a concentration story.	Top 5 Vehicle Types by Ride Distance

```
Average Ride Distance = AVERAGE ( Fact_Rides[Ride Distance (km)] )

Distance Share %      = DIVIDE (
                        [Total Ride Distance],
                        CALCULATE ( [Total Ride Distance], ALLSELECTED () )
                        )
```

E Cancellation

5 MEASURES

All five share Total Bookings as the denominator, so together with Completion Rate they partition 100% exactly.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS
Customer Cancellation Rate	7.0%	Rider-initiated failure rate. Pairs with the customer-reason pie to say how big the problem is, not just what it is made of.	Cancelled Reasons (Customer) header
Driver Cancellation Rate	18.0%	Driver-initiated failure rate — 2.6× the customer rate, and the biggest single lever on the dashboard.	Cancelled Reasons (Driver) header
Cancellation Rate	25.0%	Both sides combined. The headline cancellation number.	KPI card
No Driver Found Rate	7.0%	Unmet demand rate. Kept out of Cancellation Rate on purpose — no one cancelled, supply simply was not there.	Overall Performance
Incomplete Ride Rate	6.0%	Rate of billed trips that failed mid-journey. The service-quality tail.	Overall Performance

62.0 + 7.0 + 18.0 + 7.0 + 6.0 = 100.0% — verified, no gaps and no overlaps.

```
Customer Cancellation Rate = DIVIDE ( [Cancelled by Customer], [Total Bookings] )
Driver Cancellation Rate   = DIVIDE ( [Cancelled by Driver],   [Total Bookings] )
Cancellation Rate          = DIVIDE ( [Cancellation Count],   [Total Bookings] )
No Driver Found Rate       = DIVIDE ( [No Driver Found],       [Total Bookings] )
Incomplete Ride Rate       = DIVIDE ( [Incomplete Rides],      [Total Bookings] )
```

F Customer

3 MEASURES

Deliberately thin. With 148,788 distinct customers across 150,000 rides, this dataset has almost no repeat behaviour to analyse — and these three measures are what prove it.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT
Unique Customers	148,788	Distinct riders. DISTINCTCOUNT is right here because Customer ID repeats meaningfully. Non-additive — never total it across a date axis.	KPI card; T Customer:
Average Rides per Customer	1.008	Repeat-usage intensity. At 1.008 it is the number that justified refusing a Customer dimension.	Revenue p footnote
Revenue per Customer	₹348.46	Gross value per <i>acquired</i> customer. Read the caveat below before putting this on a card.	Revenue p

```
Unique Customers          = DISTINCTCOUNT ( Fact_Rides[Customer ID] )
Average Rides per Customer = DIVIDE ( [Total Bookings], [Unique Customers] )
Revenue per Customer      = DIVIDE ( [Total Booking Value], [Unique Customers] )
```

CAVEAT ON REVENUE PER CUSTOMER

The denominator counts *every* customer, including the ones whose only booking was cancelled or never matched. So ₹348.46 is revenue per acquired customer, not revenue per paying customer. That is a legitimate and useful metric — it is effectively a yield-on-demand figure — but it must be labelled as

such, or a reader will assume it means average customer spend. If you want spend per payer instead, filter the visual to Status Group = Successful or Failed and the same measure returns ₹508 territory.

G Vehicle

0 MEASURES — BY DESIGN

WHY THIS GROUP IS EMPTY

Your brief asked for a Vehicle measure group. Having built the model, **no vehicle-specific measure is needed**, and adding one would violate your own instruction not to create unnecessary measures.

Every question the Vehicle Type view asks is answered by taking an existing measure and putting Dim_Vehicle[Vehicle Type] on the axis. Top 5 by distance is Total Ride Distance with a Top-N filter. Ratings by vehicle is AverageDriver Rating sliced by the same column. Concentration is Distance Share % , which is generic and already lives in group D.

A measure named *Distance by Vehicle Type* would be pure duplication — the star schema is what makes it unnecessary. This is the payoff for building the dimension rather than a flat table.

H Rating

4 MEASURES

Ratings exist only on the 93,000 completed rides. AVERAGE ignores blanks, so all four are automatically scoped to rated rides with no filter written anywhere.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS USED
Average Driver Rating	4.23	Mean score riders gave drivers, over completed rides. Service quality from the rider's side.	Driver Ratings Distribution cards, by vehicle
Average Customer Rating	4.40	Mean score drivers gave riders. Quality from the supply side.	Customer Rating Analysis cards, vehicle
Rated Rides	93,000	The base the two averages are computed over. Put it beside the cards so nobody assumes 4.23 is an average of all 150,000	Ratings page footnote

bookings.			
Rating Gap	+0.17	Customer minus driver. Riders are consistently harder markers than drivers — a real, if small, asymmetry that the two separate cards would never surface on their own.	Ratings page insight tile

```
Average Driver Rating  = AVERAGE ( Fact_Rides[Driver Rating] )
Average Customer Rating = AVERAGE ( Fact_Rides[Customer Rating] )
Rated Rides            = COUNT ( Fact_Rides[Driver Rating] )
Rating Gap             = [Average Customer Rating] - [Average Driver Rating]
```

Time Intelligence

7 MEASURES

All seven work on this dataset. Year-over-year measures are deliberately absent — see the omissions section.

MEASURE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT IS USED
Rides YTD	Cumulative bookings from 1 January. Reached 150,000 by 30 December; makes pace visible on a running line.	Ride Volume Over Time, secondary line
Revenue YTD	Cumulative billed value. The executive's running total.	Revenue trend
Rides Previous Month	Same measure shifted back one month. The comparison basis, useful as a faint reference line.	Monthly volume chart
Rides MoM %	Month-on-month growth. Ranged -7.3% (Feb) to +6.6% (Mar) — genuine seasonal movement worth showing.	KPI card with conditional colour
Revenue Previous Month	Revenue comparison basis.	Revenue trend
Revenue MoM %	Revenue growth. Diverges from ride growth — March was +6.6% rides but +11.8% revenue, meaning fare mix improved, not just volume.	KPI card
Rides Rolling 7-Day Avg	Trailing seven-day mean, ~411 rides/day. Strips weekday noise out of the daily line so the trend is readable.	Ride Volume Over Time, smoothing line

Rolling 7-Day Avg is designed for a daily axis. On a monthly axis it returns the trailing window at month end, which is not what a reader will assume — keep it on the daily chart.

```
Rides YTD = TOTALYTD ( [Total Bookings], Dim_Date[Date] )
Revenue YTD = TOTALYTD ( [Total Booking Value], Dim_Date[Date] )

Rides Previous Month = CALCULATE ( [Total Bookings],
                                   DATEADD ( Dim_Date[Date], -1, MONTH ) )
Rides MoM % = DIVIDE ( [Total Bookings] - [Rides Previous Month],
                       [Rides Previous Month] )

Revenue Previous Month = CALCULATE ( [Total Booking Value],
                                    DATEADD ( Dim_Date[Date], -1, MONTH ) )
Revenue MoM % = DIVIDE ( [Total Booking Value] - [Revenue Previous Month],
                        [Revenue Previous Month] )

Rides Rolling 7-Day Avg = AVERAGEX (
    DATESINPERIOD ( Dim_Date[Date], MAX ( Dim_Date[Date]
    [Total Bookings]
    )
```

All seven depend on Dim_Date being marked as the date table, which it is. Without that marking TOTALYTD and DATEADD would return wrong results silently rather than erroring.

J Operational Quality

3 + 5 QA MEASURES

Service-level metrics, plus the five regression tests from the audit in a QA subfolder.

MEASURE	VALUE	WHAT IT CALCULATES & WHY IT MATTERS	WHERE IT
Average Pickup Wait (min)	8.46	Mean minutes from booking to pickup, over the 139,500 rides where a driver was allocated. The rider's first impression.	Operations by vehicle;
Average Trip Duration (min)	29.15	Mean in-vehicle minutes over billed rides. Pairs with distance to expose congestion.	Operations by daypart
Driver Allocation Rate	93.0%	Share of bookings that found a driver at all. Isolates pure supply failure from	Overall Performan

Ride Booking Measure Layer
cancellation behaviour — these are
different problems with different fixes.

QA × 5	all 0	Standing regression tests from the audit: out-of-range ratings, non-positive values, ratings on unrated statuses, revenue without a payment method, quoted identifiers.	Hidden QA
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```
Average Pickup Wait (min) = AVERAGE ( Fact_Rides[Avg Pickup Wait (min)] )
Average Trip Duration (min) = AVERAGE ( Fact_Rides[Avg Trip Duration (min)] )
Driver Allocation Rate = DIVIDE ( [Operational Rides], [Total Bookings] )
```

VALIDATION **KPI-by-KPI reconciliation**

Every measure below was executed against the model and compared with a value computed independently from the raw CSV in Python. Not one figure was taken on trust.

Total Bookings 150,000		RECONCILED
INCLUDED	All 150,000 rows, every status. Booking attempts, not completed trips.	
EXCLUDED	Nothing.	
FILTER CONTEXT	Correct. COUNTRROWS on the fact honours every incoming filter with no override.	
DOUBLE-COUNTING	Impossible. DISTINCTCOUNT(Ride Key) = 150,000 = row count, so the grain is exactly one row per booking.	
TRAP AVOIDED	DISTINCTCOUNT(Booking ID) would return 148,767 and lose 1,233 real rides.	

Completed Rides 93,000		RECONCILED
INCLUDED	Booking Status = Completed only.	
EXCLUDED	57,000 rows: 27,000 driver-cancelled, 10,500 customer-cancelled, 10,500 unmatched, 9,000 incomplete.	

FILTER CONTEXT	Correct, and tested. Under Vehicle Type = Auto the five status measures sum to 37,419, which equals Total Bookings for Auto. KEEPFILTERS is doing its job.
DOUBLE-COUNTING	None. The five statuses are mutually exclusive and exhaustive; their sum is exactly 150,000.

Total Booking Value ₹51,846,183		RECONCILED
INCLUDED	102,000 rows — 93,000 Completed (₹47,260,574) plus 9,000 Incomplete (₹4,585,609). The two sum to the total exactly.	
EXCLUDED	48,000 rows, automatically. Cancelled and unmatched bookings have a null Booking Value and SUM ignores nulls.	
FILTER CONTEXT	Correct, and notably no CALCULATE filter is required . Writing CALCULATE(SUM(...), Status = "Completed") here would not exclude cancellations — they are already excluded by the data — it would exclude <i>incomplete</i> revenue, which is a different and wrong decision.	
CANCELLED RIDES	Must not contribute . No trip, no fare. Verified: ₹0 across all 37,500.	
INCOMPLETE RIDES	Must contribute . All 9,000 carry a fare, a distance and a payment method — they are billed trips that ended badly. Excluding them understates revenue by ₹4,585,609, or 8.8%.	
NO DRIVER FOUND	Must not contribute . Nothing was dispatched or billed. Verified: ₹0 across all 10,500.	

Completed Ride Revenue ₹47,260,574		RECONCILED
INCLUDED	The 93,000 completed rides only.	
EXCLUDED	The 9,000 incomplete rides worth ₹4,585,609, plus all zero-value rows.	
WHY BOTH EXIST	Total Booking Value is what was billed; Completed Ride Revenue is what was billed on trips that worked. The gap between them <i>is</i> the incomplete-ride problem, quantified. Showing only one hides it.	

Completion Rate 62.0%		RECONCILED — NOTE THE DENOMINATOR
CALCULATION	93,000 ÷ 150,000. The denominator is <i>all booking attempts</i> , including the 10,500 that never found a driver.	
IS THAT RIGHT?	Yes, for a customer-experience KPI — from the rider's side an unmatched request is a failure like any other. But it is a choice, and	

	it should be labelled.
THE ALTERNATIVE	A driver-performance view would use <code>Operational Rides</code> (139,500) as the denominator, giving 66.7%. Rather than build a second competing rate, the model exposes <code>Driver Allocation Rate</code> (93.0%) so the supply failure is visible as its own number.
DOUBLE-COUNTING	None. Numerator is a strict subset of the denominator.

Cancellation Rates 7.0% / 18.0% / 25.0%		RECONCILED
INCLUDED	Customer 10,500; Driver 27,000; combined 37,500. All over 150,000 attempts.	
EXCLUDED	No Driver Found is deliberately not counted as a cancellation. Nobody cancelled — supply failed. Folding it in would report 32% cancellation and overstate the behavioural problem by seven points.	
DOUBLE-COUNTING	None. Customer + Driver = Cancellation Count exactly (10,500 + 27,000 = 37,500), and the two statuses cannot co-occur on a row.	
PARTITION CHECK	62.0 + 7.0 + 18.0 + 7.0 + 6.0 = 100.0%. Verified.	

Average Booking Value ₹508.30		RECONCILED — NAIVE VERSION REJECTED
INCLUDED	The 102,000 billed rides. <code>AVERAGE</code> skips blanks, so the denominator is right with no filter written.	
CROSS-CHECK	₹51,846,183 ÷ 102,000 = ₹508.2959. Matches to six decimals.	
REJECTED ALTERNATIVE	<code>DIVIDE([Total Booking Value], [Total Bookings])</code> returns ₹345.64 — billed revenue spread across 48,000 bookings that were never billed. It is not an average fare of anything and would have been on the dashboard as one.	

Average Ride Distance 24.64 km		RECONCILED — NAIVE VERSION REJECTED
INCLUDED	The same 102,000 rides that have a distance recorded.	
CROSS-CHECK	2,512,975.19 ÷ 102,000 = 24.637012. Exact match.	
REJECTED ALTERNATIVE	Dividing by all 150,000 bookings gives 16.75 km , a 32% understatement of the typical trip.	

Average Driver Rating 4.23		RECONCILED
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INCLUDED	93,000 completed rides. Scoping happens automatically because ratings are null on every other status.
EXCLUDED	57,000 unrated bookings — correctly, since no rating was ever given.
FILTER CONTEXT	Correct with no <code>CALCULATE</code> at all. This is the direct payoff of refusing to fill nulls in Step 01: had the pipeline replaced null ratings with 0, this measure would read 2.62 and every quality card on the dashboard would be wrong.
PRESENTATION	Show <code>Rated Rides</code> (93,000) alongside, so the base is never assumed to be 150,000.

Average Customer Rating 4.40		RECONCILED
INCLUDED	The same 93,000 completed rides. Exact match to 4.404584.	
INSIGHT	Riders average 4.23 from customers while customers average 4.40 from drivers — a +0.17 gap, captured by <code>Rating Gap</code> . It is stable across all seven vehicle types (4.23–4.24 driver, 4.40–4.41 customer), which means it is a rating-behaviour asymmetry, not a vehicle quality signal.	